MCO-1D ORNITHOPTER

Commissioning Manual — Revision 17 — July 2026

Complete guide from assembly through Discovery flights. Reflects ODrive S1 architecture (8 boards × 1 motor each), full S1 health monitoring (pre-flight, in-flight, post-flight), wing morphing programme H-33 through H-36, and all updated tooling. Do not skip steps. Every step is numbered.

Aircraft	MCO-1D metachronal cascade ornithopter
Mission	Search and Rescue testbed, biomimetic research
MTOW	3.448 kg (33.8 N) — measure actual after build
Stall speed	4.85 m/s (9.4 kt) out of ground effect
VNE	28.0 m/s (54.4 kt)
Hover freq (est)	2.5 Hz — measured precisely at TC-02
Freq limit	3.0 Hz hard cap — cannot be overridden
Wings	8 — four tandem pairs W1 (front) to W4 (rear)
Motors	8 × GoolRC 5010-14 360 KV, one per wing
Motor controllers	8 × ODrive S1 — ONE BOARD PER MOTOR
Encoder	On-board absolute magnetic — no index search at boot
Brake resistors	8 × 2Ω 50W — one per S1, cannot be shared
Pitot airspeed	Pixhawk MS4525DO → V_ms — required for all morphing
Health monitoring	Pre/in/post-flight: FET temp, motor temp, current, rope wear
Wave Library	48,097 lines — 33 hypotheses, 180 variants, 148 GS cmds
Total code	65,336 lines across 31 Python files
Revision	17 — July 2026 — ODrive S1 + health monitoring

TABLE OF CONTENTS

PART I — WHAT THIS AIRCRAFT IS AND HOW IT WORKS

1. What the MCO-1D Is
2. Eight Wings and Metachronal Cascade
3. Per-Wing Sensing — What the Computer Knows
4. Wing Morphing Programme — H-33 to H-36
5. ODrive S1 Architecture — Key Change from v3.6
6. Motor Health Monitoring — Three Phases
7. Safety Rules

PART II — HARDWARE — EVERY COMPONENT

8. Wing Assemblies — Motor, Rope, Sector Wheel
9. ODrive S1 Controllers — 8 Boards × 1 Motor
10. Encoder Magnets — Installation and Verification
11. Brake Resistors — One Per Board, Cannot Share
12. Pitch Servos and AoA Sensing
13. Sweep Servos — Fore-Aft, Dynamic Sweep, Stow
14. ADC Chips — Per-Wing Feedback
15. Pitot Tube — Airspeed for Morphing
16. Load Cell — Commissioning Bench Instrument
17. Flight Computer and Sensors
18. Power System
19. Radio and Navigation
20. Wiring Reference — GPIO, SPI, CAN, I2C

PART III — ASSEMBLY — 24 STEPS

PART IV — SOFTWARE INSTALLATION — 9 STEPS

PART V — PRE-FIRST-FLIGHT SEQUENCE

21. Step 1 — Load Cell Calibration
22. Step 2 — Build Measurements
23. Step 3 — ODrive S1 Configuration
24. Step 4 — Encoder Magnet Verification
25. Step 5 — RC Channel Verification
26. Step 6 — Pre-Flight Checks (19 checks)
27. Step 7 — Vibration Baseline

PART VI — COMMISSIONING TESTS — TC-01 THROUGH TC-04

28. TC-01 Bench Test — 16 Steps
29. TC-02 Captive Hover — 8 Steps
30. TC-03 Truck-Mount Forward Flight — 6 Steps
31. TC-04 First Free Flight — 7 Steps

PART VII — FLIGHT OPERATIONS

32. Pre-Flight Checklist (19 checks)
33. Normal Flight Procedures
34. Emergency Procedures
35. Post-Flight Procedure and Health Check

PART VIII — MOTOR HEALTH MONITORING

36. Pre-Flight S1 Health Checks (5 checks)
37. In-Flight S1 Health Monitor — 6 Monitors
38. Post-Flight S1 Health Check — 5 Checks
39. Health Alert Thresholds
40. Pilot Alert Path — Panel and Ground Station

PART IX — WING MORPHING PROGRAMME

41. Overview and Pitot Airspeed Requirement
42. H-33 Variable Wing Area — Stow and Stop
43. H-34 Dynamic Sweep Kinematics
44. H-35 Thrust/Lift Role Separation
45. H-36 Minimum Power Loiter and Ferry
46. Morphing Speed Discovery Mode
47. Loiter Endurance Discovery Mode

PART X — DISCOVERY PROGRAMME

48. All 33 Hypotheses H-01 to H-36

PART XI — REFERENCE

49. GPIO and Bus Pin Reference
50. ODrive S1 Node Assignment
51. Software File Reference
52. Ground Station Commands
53. Calibration Constants Pending TC-02
54. Pre-First-Flight Checklist Summary

PART I — WHAT THIS AIRCRAFT IS AND HOW IT WORKS

SECTION 1 — WHAT THE MCO-1D IS

The MCO-1D is a prototype aircraft that flies by flapping its wings. It has eight wings in four tandem pairs. W1 is the front pair, W4 the rear. Each pair has a port (left) and starboard (right) wing — eight wings total, all completely independent. Every wing has its own motor and its own ODrive S1 motor controller. The computer commands all eight motors fifty times per second and can change what any individual wing is doing within 20 milliseconds.

SECTION 2 — METACHRONAL CASCADE

The wings do not all flap at the same time. They flap in a staggered sequence: W1 starts, then W2 begins a fraction of a cycle later, then W3, then W4. Each rearward pair captures energy from the wake of the pair ahead. Research on plesiosaurs showed correctly phased hind flippers generated 60% more thrust and 40% better efficiency. The MCO-1D uses the same principle with four pairs.

SECTION 3 — PER-WING SENSING

Sensor	Source and Rate
Angle of Attack	MCP3208 #1 (SPI1 CE1) reads pitch servo feedback pot. Actual AoA per wing at 0.044° resolution, 50 Hz.
Sweep position	MCP3208 #2 (SPI1 CE0) reads sweep servo feedback pot. Actual fore-aft per wing, 50 Hz.
Stroke position	ODrive S1 pos_estimate via CAN. Absolute — on-board encoder knows position at power-on, no index search.
Pitot airspeed	Pixhawk MS4525DO → msg.airspeed → V_ms. Used by H-33/34/35/36. NOT GPS ground speed.
FET temperature	S1 on-board thermistor → read_temperatures() at 5 Hz. Real measured value, not an estimate.
Motor temperature	S1 on-board motor thermistor. Real measured value.
Motor current	S1 Iq_measured via CAN. Used for rope wear monitoring.
Axis error state	S1 heartbeat error register, 1 Hz. Per-board fault isolation.

SECTION 4 — WING MORPHING PROGRAMME

Any wing pair can independently be in one of three states:

State	Name	Key Property
State A	Active flapping	Generates lift and thrust via unsteady aerodynamics.
State B	Stopped at neutral	Fixed-wing lift = ½ρV²S·CL. Zero motor power. At 8 m/s: two stopped pairs cover MTOW entirely.
State C	Stowed along fuselage	Sweep to 85-95° aft. Near-zero wing area. Near-zero gust loading. Full MTOW on active pairs.

Four hypotheses implement the programme — all using pitot airspeed:

Hyp	Name	Purpose
H-33	Variable wing area (18v)	Stow/stop any pair. Active pairs retrim via sweep.

H-34	Dynamic sweep (16v)	±20-40° per stroke. Forward downstroke, aft upstroke.
H-35	Thrust/lift separation (14v)	Stopped pairs=wings. Active pairs=engines.
H-36	Loiter and ferry (8v)	3 pairs stopped. 1 pair at minimum freq. ~3× endurance.

SECTION 5 — ODRIVE S1 ARCHITECTURE

■ WARNING: The MCO-1D uses 8 × ODrive S1 boards — ONE BOARD PER MOTOR. This is not the same as the v3.6 which had two motors per board. Do not attempt to connect two motors to one S1 board.

CAN node	Wing
Node 0	W1 port
Node 1	W1 stbd
Node 2	W2 port
Node 3	W2 stbd
Node 4	W3 port
Node 5	W3 stbd
Node 6	W4 port
Node 7	W4 stbd

wing_idx == node_id. No axis0/axis1 distinction. arb_id = (node_id << 5) | cmd_id for all wings. The S1 on-board absolute encoder means wing position is known at power-on — no index search, no calibration move required.

SECTION 6 — MOTOR HEALTH MONITORING — THREE PHASES

The ODrive S1 reports real FET temperatures, motor temperatures, motor currents, and axis error states over CAN. These are actual measured values — not estimates from current squared times resistance. Health monitoring runs across three phases:

Phase	When	What is checked
PRE-FLIGHT	5 S1-specific checks before arming	Heartbeats, encoder magnets, temperatures near ambient, axis errors clear, current baseline at 5 rad/s.
IN-FLIGHT	S1InFlightMonitor at 50 Hz	FET temp (warn 70°C, crit 90°C), motor temp (warn 60°C, crit 75°C), current (warn 18A, crit 22A), rope wear (>20% current rise from baseline), port/stbd asymmetry (>25%), axis errors.
POST-FLIGHT	check_s1_post_flight() at shutdown	Peak temperature review, rope trend, cooling state, error states, next-flight readiness GO/NO-GO. Saved to /home/pi/knowledge/last_flight_health.json.

SECTION 7 — SAFETY RULES

■ WARNING: Frequency ceiling 3.0 Hz, amplitude ceiling 96°, and AoA stall protection cannot be overridden from any command.

■ WARNING: H-33 State C (stow) and H-35/36 two-pair configurations are gated on TC-03 CT-J curve. Do not attempt before TC-03 completes.

■ WARNING: All aerodynamic decisions use pitot airspeed (V_ms = msg.airspeed). GPS ground speed is never used for any aerodynamic calculation.

✓ H-33/34/35/36 State B (stopped at neutral) variants are safe for TC-04 because stopped wings still generate fixed-wing lift.

✓ If the panel margin bar shows CRITICAL (red): a motor has reached a critical temperature or error state. Land as soon as practicable. The software has already begun reducing vel_limit on affected wings.

PART II — HARDWARE: EVERY COMPONENT

SECTIONS 8-11 — MOTORS AND ODRIVES S1

Motor model	GoolRC 5010-14, 360 KV, 7 pole pairs (12N14P)
Gear ratio	7:1 — sector wheel turns once per 7 motor turns
Sector arc	100° mechanical. Software limit 96°.
Rope drives	Independent per wing. No shared rope between port and stbd.
Rope wear	Rising current at same frequency = early rope wear indicator.

ODRIVE S1 NODE ASSIGNMENT

Board / Node	Wing	CAN arb_id formula	Notes
0	W1 port	(0 << 5) | cmd_id	Single axis. No axis0/axis1.
1	W1 stbd	(1 << 5) | cmd_id	
2	W2 port	(2 << 5) | cmd_id	
3	W2 stbd	(3 << 5) | cmd_id	
4	W3 port	(4 << 5) | cmd_id	
5	W3 stbd	(5 << 5) | cmd_id	
6	W4 port	(6 << 5) | cmd_id	
7	W4 stbd	(7 << 5) | cmd_id	

● CAUTION: vel_limit must be 150 rad/s on ALL 8 S1 boards. Run mco_odrive_setup.py. Factory default is too low.

SECTION 10 — ENCODER MAGNET INSTALLATION

Each S1 includes a 1/4 inch (6.35 mm) diametrically magnetised disc magnet. This replaces the external incremental encoder required by the v3.6.

Step	Action	Expected / Notes
1	Ensure motor is unpowered	Disconnect ESC/S1 before touching motor.
2	Press magnet onto shaft centre	Flush with the back face of the motor. Diametric magnetisation — the poles are across the flat faces.
3	Align S1 encoder chip over magnet	Encoder chip is on the back of the S1 PCB. Position S1 so chip is directly over magnet centre. Target gap: 1-2 mm.
4	Secure with adhesive if needed	Magnet should press-fit. If loose, use cyanoacrylate on shaft only.
5	Verify with check-magnet command	python3 mco_odrive_setup.py --check-magnet Status 1 = OK. Status 0 = magnet missing. Status 2 = too close (increase gap). Status 3 = too far (decrease gap).

■ WARNING: A missing or misaligned magnet = no position feedback = the wing cannot be controlled. check_s1_encoder_magnets() in pre-flight checks will block arming if any magnet status is not 1.

SECTION 11 — BRAKE RESISTORS

■ WARNING: Do NOT connect multiple S1 boards to a shared brake resistor. Each S1 has its own isolated DC bus. Sharing brake resistors will cause current to flow between boards and damage them.

Each S1 comes with a 2Ω 50W brake resistor. Connect it to the brake resistor terminal on that board. At stroke reversal, regenerated energy is dissipated locally — there is no bus-sharing between S1 boards.

SECTIONS 12-13 — SERVOS AND ADCs

Pitch GPIO	W1p=18,W1s=23,W2p=24,W2s=25,W3p=12,W3s=16,W4p=20,W4s=21
Sweep GPIO	W1p=4,W1s=17,W2p=27,W2s=22,W3p=5,W3s=6,W4p=13,W4s=19
Pitch ADC	MCP3208 #1 SPI1 CE1 (GPIO7) — CH0-CH7
Sweep ADC	MCP3208 #2 SPI1 CE0 (GPIO8) — CH0-CH7
Resistors	1kΩ in-line on each of 16 feedback lines
SPI1	CLK=GPIO11, MOSI=GPIO10, MISO=GPIO9
/boot/config	dtoverlay=spi1-2cs

SECTION 15 — PITOT TUBE

■ WARNING: The pitot tube is not optional. H-33/34/35/36 all require pitot airspeed. GPS ground speed is never used for aerodynamic decisions.

Sensor	Pixhawk MS4525DO digital airspeed sensor
Mount	Forward fuselage in clean undisturbed airflow
Connect	Pixhawk airspeed I2C port
Data path	Pixhawk → MAVLink VFR_HUD → mco_mavlink.py → sensors[V_ms]
Used by	H-33 stow trigger, H-34 sweep schedule, H-35 role switching, H-36 free lift calculations, morphing_speed_score()

SECTION 16 — LOAD CELL (BENCH ONLY)

Capacity	50N minimum. 100N preferred.
HX711	DT=GPIO26, SCK=GPIO7 (no conflict with SPI1 CE1 — bench only)
Software	mco_loadcell.py — calibrate before TC-01 with 1 kg mass
TC-01 use	Thrust curve: force vs frequency
TC-02 use	Hover detection: reads zero when lift = weight

SECTION 20 — WIRING REFERENCE

Pin	Function
GPIO 4	DS18B20 OAT (One-Wire)
GPIO 5	W3p sweep PWM
GPIO 6	W3s sweep PWM
GPIO 7	SPI1 CE1 pitch ADC / HX711 SCK bench
GPIO 8	SPI1 CE0 sweep ADC
GPIO 9-11	SPI1 MISO/MOSI/CLK — both ADCs
GPIO 12	W3p pitch PWM
GPIO 13	W4p sweep PWM
GPIO 16	W3s pitch PWM
GPIO 17	W1s sweep PWM
GPIO 18	W1p pitch PWM
GPIO 19	W4s sweep PWM
GPIO 20	W4p pitch PWM
GPIO 21	W4s pitch PWM
GPIO 22	W2s sweep PWM
GPIO 23	W1s pitch PWM
GPIO 24	W2p pitch PWM
GPIO 25	W2s pitch PWM / CAN adapter interrupt
GPIO 26	HX711 DT — load cell data (bench only)
GPIO 27	W2p sweep PWM
I2C1 GPIO2/3	IMU 0x68 + barometer 0x76
SPI0 CE0	MCP2515/2518FD CAN adapter
UART GPIO14/15	RC SBUS + Pixhawk MAVLink 921,600 baud

PART III — ASSEMBLY

Step	Action	Expected / Notes
1	Mount airframe	Rigid. No flex between motor mounts.
2	Mount 8 sector wheels	Full arc smooth, no binding.
3	Mount 8 motors	Loctite 243 on all fasteners. Verify 7:1 ratio.
4-5	Install rope drives and wings	One rope per wing, independent.
6	Set pitch servo neutrals — BEFORE INSTALLING	Command all to 0°. Attach horn.
7	Set sweep servo neutrals	Command to 60°. Attach horn. Command to 95° and measure actual angle (stow feasibility).
8-9	Install 8 pitch and 8 sweep servos	Route feedback wires.
10	Mount 8 ODrive S1 boards	One per wing. Vibration isolation. Wire: node 0 → W1p motor, node 1 → W1s motor, etc.
11	Install encoder magnets	Press 1/4" magnet onto each motor shaft centre. Align S1 encoder chip 1-2 mm above magnet.
12	Connect brake resistors	2Ω 50W included with each S1. Connect to brake terminal. One per board — do NOT share.
13	Wire CAN bus	Pi adapter → node 0 → 1 → 2 → 3 → 4 → 5 → 6 → 7. 120Ω terminator at Pi end AND at node 7 end.
14-15	Wire pitch and sweep servo PWM	Per Section 20 GPIO table.
16-17	Wire feedback ADC lines	Each pitch feedback → 1kΩ → MCP3208 #1. Each sweep feedback → 1kΩ → MCP3208 #2.
18	Mount Pi 4 and sensors	Vibration isolate Pi. I2C1 for IMU and baro.
19	Mount and connect pitot tube	Forward of all wings. Clean airflow. → Pixhawk airspeed port.
20	Route power cables — DO NOT CONNECT BATTERIES YET	Main battery → 30A fuse → S1 boards (each gets its own connection). Servo battery → servo rail. Isolated BEC → Pi.
21	Install RC receiver	SBUS → GPIO15. Bind to transmitter.
22	Connect Pixhawk	TELEM2 → Pi UART 921,600 baud.
23	First power-on — Pi only	Boot. SSH. python3 --version → 3.8+.
24	Complete software installation	See Part IV before proceeding further.

PART IV — SOFTWARE INSTALLATION

```
# Step 1 — Install Raspberry Pi OS Lite 64-bit
# Use Raspberry Pi Imager. Device: Pi 4. OS: Lite 64-bit. Enable SSH.

# Step 2 — /boot/config.txt additions
dtoverlay=mcp2515-can0,oscillator=8000000,interrupt=25
dtoverlay=spi1-2cs
dtoverlay=w1-gpio,gpiopin=4
dtparam=i2c_arm=on
dtparam=spi=on

# Step 3 — Install Python packages
pip install pymavlink python-can pigpio spidev smbus --break-system-packages
sudo systemctl enable pigpiod && sudo systemctl start pigpiod

# Step 4 — Copy all software files to Pi
scp mco_wave_library.py mco_main.py mco_odrive_interface.py \
mco_odrive_setup.py mco_loadcell.py mco_build_measurements.py \
mco_preflight_checks.py mco_preflight.py mco_rc_verify.py \
mco_vibration_check.py mco_test_cards.py mco_log_analysis.py \
mco_logger.py mco_mavlink.py mco_gs_server.py mco_panel.py \
mco_ground_station.py mco_can_setup.sh mco_install.sh \
pi@mco-1d.local:/home/pi/
ssh pi@mco-1d.local
mkdir -p /home/pi/logs /home/pi/knowledge
bash /home/pi/mco_install.sh

# Step 5 — CAN bus
sudo reboot && ssh pi@mco-1d.local
sudo bash mco_can_setup.sh
ip link show can0 # must show UP

# Step 6 — Configure ODrive S1 boards (apply main battery power first)
python3 mco_odrive_setup.py
# Configures all 8 nodes: vel_limit=150, pole_pairs=7, velocity mode

# Step 7 — Verify encoder magnets
python3 mco_odrive_setup.py --check-magnet
# Status 1 = OK for all 8. 0=missing, 2=too close, 3=too far.

# Step 8 — Motor calibration
python3 mco_odrive_setup.py --calibrate
# Motor resistance + inductance only. No encoder calibration needed.

# Step 9 — Set initial cascade phases
# from mco_odrive_interface import ODriveInterface
# odi = ODriveInterface(); odi.enable_all()
```

```
# odi.set_initial_phases([0,45,90,135,180,225,270,315])
# Adjust values per actual cascade phase offsets in wave library.
```

PART V — PRE-FIRST-FLIGHT SEQUENCE

Run these tools in order before TC-01. Do not skip any step.

Step	Action	Expected / Notes
21a	Load cell calibration	python3 mco_loadcell.py --calibrate Place 1.000 kg mass on cell. Saves to /home/pi/knowledge/loadcell_cal.json. REQUIRED before TC-01. Cannot proceed without it.
21b	Load cell test	python3 mco_loadcell.py Live readout. Verify reading changes with applied force.
22	Build measurements	python3 mco_build_measurements.py Measure 15 physical dimensions with tape measure and kitchen scale. Takes ~30 min. Patches wave library with real values. REQUIRED before TC-01 for accurate aerodynamic calculations.
23	ODrive S1 configuration	python3 mco_odrive_setup.py All 8 boards. vel_limit=150. pole_pairs=7. Saves to non-volatile.
24	Encoder magnet verification	python3 mco_odrive_setup.py --check-magnet All 8 nodes must return status 1. Status 0 = missing magnet → install before proceeding.
25	RC channel verification	python3 mco_rc_verify.py Verify CH1=Roll, CH2=Pitch, CH3=Lever, CH4=Yaw. Saves to /home/pi/knowledge/rc_mapping.json.
26	Pre-flight checks — all 19	python3 mco_preflight_checks.py --bench Includes 5 S1-specific checks: check_s1_heartbeats — all 8 boards heartbeating check_s1_encoder_magnets — all 8 magnets status 1 check_s1_temperatures_preflight — within 20°C of ambient check_s1_axis_errors_preflight — all errors clear check_s1_current_baseline — rope/bearing health at 5 rad/s All 19 checks must return PASS or WARN. Any FAIL blocks arming.
27	Vibration baseline	python3 mco_vibration_check.py --quick IMU and SPI baseline before any motors run. Run full sweep after TC-01-06 (motor spin).

PART VI — COMMISSIONING TESTS

Four test cards. Never advance to the next until the current passes fully. S1 health monitoring is active throughout all test cards.

Card	Arrangement	Notes
TC-01	Bench (post) load cell	16 steps. 10 auto, 6 semi-auto. No forward motion.
TC-02	Same post, same cell	8 steps. Fully auto. Finds hover frequency.
TC-03	Truck-mount	6 steps. Fully auto. Truck provides airspeed. Maps CT-J.
TC-04	First free flight	7 steps. Guided auto. Pilot in command.

SECTION 28 — TC-01 BENCH TEST: 16 STEPS

Start: tc01_start Monitor: tc01_status Confirm: tc01_confirm

Step	Action	Expected / Notes
01	Servo neutral check — AUTO	All 16 servos to neutral. ADC reads. Pass: all within ±2°.
02	Servo range ±45° — AUTO	Commands ±45°. Pass: all reach range within ±3°.
03	Pitch ADC all 8 — AUTO	MCP3208 #1 all channels. Pass: all readable.
04	Sweep ADC all 8 — AUTO	MCP3208 #2 all channels. Pass: all readable.
05	ODrive S1 8-node comms — AUTO	read_all_states() on all 8 nodes. Pass: all 8 heartbeating. Also reads S1 temperatures for first baseline.
06	Motor spin at 10 rad/s — AUTO	All 8 nodes. Reads currents and S1 temperatures. Verify with IR thermometer: all below 60°C. After this step: run mco_vibration_check.py (full sweep).
07	Wing flapping 0.5 Hz 30 s — AUTO	Inspect rope drives visually.
08	Wing flapping 2.5 Hz 60 s — AUTO	S1 reads real FET and motor temps. Pass: all below 60°C.
09	Pitch ADC calibration — SEMI-AUTO	Software commands -45°, 0°, +45°. Operator visually confirms. tc01_confirm.
10	Sweep ADC calibration — SEMI-AUTO	Same for sweep servos.
11	Frequency ceiling 2.0-3.0 Hz — AUTO	S1 thermal monitoring throughout.
12	Static thrust measurement — SEMI-AUTO	Confirm load cell fixture attached. tc01_confirm. Software steps 1.5-3.0 Hz. HX711 reads force. Pass: max thrust ≥ MTOW (33.8 N).
13	RC mapping — SEMI-AUTO	Move sticks on request. Confirms CH1-4.
14	Safety caps and fault detection — SEMI-AUTO	Tests 3.0 Hz cap, 96° amp cap, fault detection on motor disconnect.
15	Port/stbd symmetry — AUTO	AoA symmetry via MCP3208 feedback.
16	Export — AUTO	Results and S1 health summary saved. Run mco_log_analysis.py --tc01.

✓ After TC-01-06: run python3 mco_vibration_check.py before TC-01-07. S1 reports real temperatures throughout TC-01 — monitor on ground station with motor_health command.

SECTION 29 — TC-02 CAPTIVE HOVER: 8 STEPS

Same post, same load cell. No forward motion. Fully automatic. Start: tc02_start Monitor: tc02_status

Step	Action	Expected / Notes
01	Load cell check — AUTO	HX711 responding. Calibration loaded.
02	Lift-frequency curve — AUTO	1.5-3.0 Hz. Load cell reads at each step. Computes dL/df slope.
03	Find hover frequency — AUTO	Steps until load cell reads ≤1N. HOVER FREQUENCY = load cell reads zero. Most important measurement.
04	Confirm hover — AUTO	Hold at hover freq 5s. Residual ≤2N. Pass.
05	AoA sweep at hover freq — AUTO	AoA 8-22°. Builds CL-alpha curve.
06	H-26 side force — AUTO	Differential AoA. Classifies: ROLL or LATERAL.
07	Calibrate controllers — AUTO	Applies hover_freq_hz and dL/df to descent controller.
08	5-minute captive discovery — AUTO	Discovery Mode on post.

SECTION 30 — TC-03 TRUCK FORWARD FLIGHT: 6 STEPS

■ WARNING: Pitot tube must be reading valid airspeed before TC-03. V_ms must show non-zero when truck is moving.

Step	Action	Expected / Notes
01	Stationary baseline — AUTO	Aircraft at hover freq. Truck stationary.
02	Transition sweep — AUTO	GS shows truck target speed. Finds minimum freq at each speed.
03	CT-J curve — AUTO	6 truck speeds. Records CT vs advance ratio J. REQUIRED for H-33 State C / H-35 2-pair / H-36 stow.
04	Speed envelope — AUTO	IMU stability at increasing truck speeds.
05	Descent simulation — AUTO	Freq below hover at cruise. Load cell reads tendency.
06	30-min discovery at cruise — AUTO	morphing_speed discovery. H-33/34/35 State B variants. morphing_speed_report shows leaderboard.

SECTION 31 — TC-04 FIRST FREE FLIGHT: 7 STEPS

Step	Action	Expected / Notes
01	Pre-flight confirmation	S1 health checks run. All calibrations from TC-02/03 confirmed.
02	First hover	Advance lever slowly. 3 clean liftoffs. Software records free-flight hover freq vs TC-02 captive.

03	Hover stability 30 s	Hold 30 s. sigma_h < 0.5 m. Pass.
04	Altitude hold — descent controller	100 fpm and 300 fpm tests.
05	Roll response	20/40/60% roll stick. Records rates and settle times.
06	Forward flight	Advance lever. Fly 100 m. Return. 3 times.
07	First free-flight discovery — 5 min	H-33/34/35 State B variants. Safe: stopped wings generate lift. motor_health command shows real-time S1 health throughout.

PART VII — FLIGHT OPERATIONS

SECTION 32 — PRE-FLIGHT CHECKLIST (19 CHECKS)

Step	Action	Expected / Notes
1	python3 mco_preflight_checks.py --flight	All 19 checks. S1 heartbeats, encoder magnets, temperatures, errors, current baseline all included. Battery interlock: <80% = NO-GO for TC.
2	Area clear	30 m zone. Landing zone identified.
3	Wind	<15 kt normal ops. <5 kt for TC-04 first flight.
4	SD space	At least 500 MB free.
5	Panel check	All 6 margin bars green/normal. Motor health bar normal. Any CRITICAL (red) = do not arm.
6	ARM	Lever at minimum. Confirm correct stick directions.

SECTION 33 — NORMAL FLIGHT

Takeoff	Advance lever smoothly to calibrated hover position.
Hover	Lever at hover position. Monitor panel and GS motor_health.
Forward	Lever 65-70%. descent_ctrl_engage 0 for altitude hold.
Landing	descent_ctrl_engage 100 for 100 fpm precision.

SECTION 34 — EMERGENCY PROCEDURES

Condition	Action
Panel shows WARNING (amber)	Monitor. Land at first convenient opportunity. vel_limit reducing on affected wing(s). Check motor_health on GS.
Panel shows CRITICAL (red)	Land as soon as practicable. Motor(s) at FET >90°C or motor >75°C. Software has idled affected wings and reduced vel_limit. Remaining wings are compensating. Land promptly.
Motor overcurrent alert	Check motor_health: rope binding or bearing failure. Land and inspect before next flight.
Rope asymmetry alert	One wing working harder than its partner. Land and inspect rope drive on named wing.
Wing failure in flight	7 wings: land when convenient. 6 wings: land within 2 min. 5 or fewer: land immediately.
All morphing disarm	h33_h34_h35_disengage_all — all pairs to normal flapping.

SECTION 35 — POST-FLIGHT PROCEDURE

Step	Action	Expected / Notes
1	DISARM	All motors stop.
2	POST-FLIGHT button	8-minute cooling period.
3	S1 post-flight check	Runs automatically at shutdown: check_s1_post_flight(). Reports: peak temps, rope trend, cooling state, errors, next-flight GO/NO-GO. Saved to /home/pi/knowledge/last_flight_health.json.
4	Rope inspection	Inspect all 8 rope drives. Fraying = replace before next flight.
5	Export	knowledge_export_usb and morphing_speed_report
6	Log analysis	python3 mco_log_analysis.py on laptop for anomaly review.

PART VIII — MOTOR HEALTH MONITORING

SECTION 36 — PRE-FLIGHT S1 HEALTH CHECKS

These five checks run as part of the standard 19-check pre-flight and use the ODrive S1's hardware reporting capabilities directly.

Check	What it does
check_s1_heartbeats	Polls all 8 S1 boards for CAN heartbeat frames. Missing heartbeat = board not powered or CAN broken. Names the specific missing wing(s). FAIL blocks arming.
check_s1_encoder_magnets	Reads field_status from on-board encoder on each board. Status 1=OK, 0=missing, 2=too close, 3=too far. Any non-1 status: named wing blocked until corrected.
check_s1_temperatures_preflight	Reads real FET and motor temps from all 8 boards. Must be within 20°C of ambient (OAT sensor) before arming. Hot motors before flight = previous damage or insufficient cool-down.
check_s1_axis_errors_preflight	Reads error register from all 8 boards. Auto-clears non-critical errors. Persistent errors = FAIL. Cleared-but-existed = WARN (investigate cause).
check_s1_current_baseline	Spins all 8 motors at 5 rad/s for 2 seconds. High current at low speed = rope binding, bearing fault, winding issue. Port/stbd asymmetry >25% = uneven rope tension. Baseline values stored for in-flight rope wear monitoring.

SECTION 37 — IN-FLIGHT S1 HEALTH MONITOR

S1InFlightMonitor runs at 50 Hz. Temperatures read at 5 Hz, error states at 1 Hz. Six monitors per wing:

Monitor	Threshold	Action
FET temperature	warn 70°C / critical 90°C	Above warn: vel_limit reduces dynamically per-wing via set_velocity_limit(). Above crit: S1 hardware cuts out. Motor idles automatically.
Motor temperature	warn 60°C / critical 75°C	Above warn: pilot alerted via panel margin bar. Above crit: land as soon as practicable.
Motor current	warn 18A / critical 22A	Above warn: check rope and bearings. Above crit: possible motor or drive fault.
Rope wear proxy	>20% current rise from baseline	Rising current at same frequency = rope stretch, wear, or tension change. Named wing alerted. Inspect rope before next flight.
Port/stbd asymmetry	>25% difference within pair	One wing of a pair working harder than the other. Check rope tension on the higher-current side.
Axis errors	any non-zero error code	Named in alert with hex error code. Auto-cleared if non-critical. Hard faults reported for investigation.

SECTION 38 — POST-FLIGHT S1 HEALTH CHECK

Runs automatically at shutdown. Produces 5-check maintenance report:

Check	What it reports

Peak temperature review	Peak FET and motor temps for each wing vs thresholds. Motor temp > 75°C = inspect winding before next flight.
Rope trend (current rise)	Peak current vs baseline per wing. >20% rise = inspect named rope drive before next flight.
Post-flight temperature	Real-time temp reading after cooling period. Motors should be within 20°C of ambient before storing or covering.
Post-flight axis errors	Errors that developed in the last moments of flight. Must be investigated before next flight.
Next-flight readiness	GO if all above pass. NO-GO if any critical finding. Report saved to /home/pi/knowledge/last_flight_health.json

SECTION 39 — HEALTH ALERT THRESHOLDS

Parameter	Threshold	Response
FET temperature warn	70°C	vel_limit begins reducing proportionally
FET temperature critical	90°C	motor idles (S1 hardware cutoff imminent)
Motor temperature warn	60°C	panel WARNING, land at opportunity
Motor temperature critical	75°C	panel CRITICAL, land soon
Current warn	18A	check rope and bearings after flight
Current critical	22A	possible motor or drive fault
Rope wear trigger	>20% current rise	inspect named rope drive
P/S asymmetry trigger	>25% within pair	inspect rope tension

SECTION 40 — PILOT ALERT PATH

Motor health alerts follow this path from S1 hardware to pilot:

Stage	What happens
S1 hardware	Real FET temp, motor temp, current, error register measured at board
read_temperatures()	CAN query at 5 Hz → sensors[motor_temps] in mco_main.py
read_all_states()	CAN query at 1 Hz → sensors[wing_errors] in mco_main.py
S1InFlightMonitor.update()	Runs every tick → result[motor_health] with per-wing data
Telemetry	motor_health, motor_alerts, motor_alert_level sent at 10 Hz
mco_panel.py	Motor alert level feeds margin bar: NORMAL/WARNING/CRITICAL
Margin bar colour	Normal=green. Warning=amber. Critical=red (overrides aero margin).
Ground station	motor_health → per-wing display. motor_peaks → worst values this flight.

PART IX — WING MORPHING PROGRAMME

SECTION 41 — OVERVIEW AND PITOT AIRSPEED REQUIREMENT

■ WARNING: All morphing decisions use PITOT AIRSPEED (V_ms = msg.airspeed from Pixhawk). GPS ground speed is never used for any aerodynamic decision.

State	Name	Property
State A	Flapping	Normal. Lift + thrust.
State B	Stopped at neutral	Fixed-wing lift = ½ρV²S·CL. Zero motor power.
State C	Stowed along fuselage	Near-zero wing area. Near-zero gust loading.

Free lift from stopped pairs (best L/D AoA, pitot airspeed):

Pitot airspeed	1 pair stopped	2 pairs stopped	3 pairs stopped
6 m/s	1 pair stopped: 7.5N (22%)	2 pairs: 15N (44%)	3 pairs: 19.7N (58%)
7 m/s	1 pair: 10.2N (30%)	2 pairs: 20.5N (61%)	3 pairs: 26.8N (79%)
8 m/s	1 pair: 13.3N (39%)	2 pairs: 26.9N (80%)	3 pairs: 35.0N (104%) ←
9 m/s	1 pair: 16.9N (50%)	2 pairs: 34.0N (101%) ←	3 pairs: 44.3N (131%)

← = stopped pairs cover full MTOW. Active pairs owe zero lift. All power to thrust.

SECTIONS 42-47 — H-33 through H-36 and Discovery Modes

Hyp	Variants	Description	TC gate
H-33	18v	Stow/stop any pair. Sweep trim on active pairs.	TC-04 State B safe. State C needs TC-03.
H-34	16v	±20-40° sweep per stroke. Stopped-pair speed-adaptive sweep.	v01/v02 TC-04 safe. v03 H-27 gated.
H-35	14v	Stopped pairs = wings. Active pairs = engines. AoA scheduled.	v05/v06 TC-04 safe. 2-pair needs TC-03.
H-36	8v	3 pairs stopped. 1 pair at minimum freq. ~3× endurance.	v03/v04 TC-04 safe. Hunt variants after stable FF.

Command	What it does
morphing_speed_start	H-33/34/35 priority. 48-variant scheduler. 45 s per trial. Speed primary.
loiter_endurance_start	H-36 priority. Power/endurance primary. Ferry range secondary.
morphing_speed_report	Leaderboard sorted by best pitot airspeed achieved.
h36_status	Power, endurance estimate, free lift at current airspeed.
h33_h34_h35_disengage_all	Return all pairs to normal 4-pair flapping.

PART X — DISCOVERY PROGRAMME

SECTION 48 — ALL 33 HYPOTHESES H-01 TO H-36

ID	Name	Variants
H-01	Dragonfly phase	5v
H-02	Bat advance rotation	5v
H-03	Manta spanwise wave	3v
H-04	Pterosaur soar	1v
H-05	Hawkmoth LEV	4v
H-06	Negative cascade	1v
H-07	Split frequency	3v
H-08	Manta Strouhal	15v
H-12	Locust reversal burst	3v
H-13	Plesiosaur phase	4v
H-14	Plesiosaur brake	1v
H-15	Plesiosaur dominance	6v
H-16	Turtle figure-eight	4v
H-17	Turtle asymmetric	1v
H-18	Turtle sweep-clap	3v
H-19	Sea lion AoA	4v
H-20	Otter advance ratio	5v
H-21	Sea lion fling	3v
H-22	Bat ventral clap	4v
H-23	Bat speed-adaptive AoA	6v
H-24	MCO-1D AoA gradient	4v
H-25	MCO-1D per-stroke roll	4v
H-26	MCO-1D force decouple	4v
H-27	Servo bandwidth	12v
H-28	Asymmetric cascade	6v
H-29	Resonance seeking	4v
H-30	Power smoothing	3v
H-31	Fourier stroke path	4v
H-32	Multi-wake phase	2v
H-33	Variable wing area (NEW R17)	18v
H-34	Dynamic sweep (NEW R17)	16v

H-35	Thrust/lift separation (NEW R17)	14v
H-36	Loiter and ferry (NEW R17)	8v

PART XI — REFERENCE

SECTION 49 — GPIO PIN REFERENCE

Pin	Function
GPIO 4	DS18B20 OAT
GPIO 5	W3p sweep
GPIO 6	W3s sweep
GPIO 7	SPI1 CE1 pitch ADC / HX711 SCK
GPIO 8	SPI1 CE0 sweep ADC
GPIO 9	SPI1 MISO
GPIO 10	SPI1 MOSI
GPIO 11	SPI1 CLK
GPIO 12	W3p pitch
GPIO 13	W4p sweep
GPIO 16	W3s pitch
GPIO 17	W1s sweep
GPIO 18	W1p pitch
GPIO 19	W4s sweep
GPIO 20	W4p pitch
GPIO 21	W4s pitch
GPIO 22	W2s sweep
GPIO 23	W1s pitch
GPIO 24	W2p pitch
GPIO 25	W2s pitch / CAN int
GPIO 26	HX711 DT (bench)
GPIO 27	W2p sweep
I2C1 GPIO2/3	IMU 0x68 + baro 0x76
SPI0 CE0	CAN adapter
UART GPIO14/15	RC SBUS + Pixhawk MAVLink

SECTION 50 — ODRIVE S1 NODE ASSIGNMENT

Node 0 = W1 port	Node 1 = W1 stbd
Node 2 = W2 port	Node 3 = W2 stbd
Node 4 = W3 port	Node 5 = W3 stbd
Node 6 = W4 port	Node 7 = W4 stbd

arb_id formula	(node_id << 5) | cmd_id — same for all wings
wing_idx == node_id	Direct 1:1 mapping. No axis0/axis1.
Absolute encoder	Position known at power-on. No index search.
Brake resistors	One 2Ω 50W per board. Included. Cannot share.

SECTION 51 — SOFTWARE FILE REFERENCE

File	Lines	Purpose
mco_wave_library.py	48,097	33 hypotheses, 180 variants, 148 GS cmds. All morphing ticks wired.
mco_odrive_interface.py	447	8 × S1. New: read_temperatures, read_all_states, set_velocity_limit.
mco_odrive_setup.py	282	--verify --calibrate --check-magnet --node N.
mco_preflight_checks.py	1,256	22 check fns. Pre/in/post-flight S1 health system.
mco_main.py	483	S1 temps 5 Hz, errors 1 Hz. Dynamic vel_limit. Post-flight check.
mco_panel.py	506	Motor health alerts on margin bar. CRITICAL overrides aero margin.
mco_loadcell.py	527	HX711. Calibrate/tare/thrust sweep/hover detect.
mco_build_measurements.py	328	15 measurements. Patches wave library.
mco_preflight.py	782	Three-tier preflight original.
mco_rc_verify.py	324	RC mapping + lever interlock.
mco_vibration_check.py	367	Vibration sweep. SPI stability. Tick budget.
mco_test_cards.py	1,665	TC-01 through TC-04.
mco_log_analysis.py	462	Post-flight analysis. Anomaly detection.
mco_ground_station.py	660	motor_health, motor_peaks, morphing commands.

SECTION 52 — SELECTED GROUND STATION COMMANDS

Command	Function
arm / disarm	Arm or disarm all 8 wings
motor_health	Per-wing FET temp, motor temp, current, active alerts
motor_peaks	Worst values recorded this flight for all 8 wings
tc01_start / tc01_confirm / tc01_status	TC-01 control
tc02_start / tc02_status	TC-02 captive hover
tc03_start / tc03_status	TC-03 truck forward flight
tc04_start / tc04_continue / tc04_status	TC-04 free flight
morphing_speed_start	H-33/34/35 speed exploration — all 48 variants
morphing_speed_report	Leaderboard by pitot airspeed achieved
h33_status / h34_status / h35_status / h36_status	Per-hypothesis status

h33_h34_h35_disengage_all	Return all pairs to normal flapping
loiter_endurance_start	H-36 loiter/ferry exploration
h36_status	Power, endurance estimate, free lift report
disc_mode_engage	Full Discovery Mode — all 12 focus modes
sensing_display	Full per-wing AoA, sweep, stroke, effective AoA
servo_health_report	Tracking RMS per wing — rope/servo wear indicator
stall_detector_status	Per-wing stall likelihood
knowledge_export_usb	Save discovery registry to USB
failure_briefing	Pre-flight failure performance table

SECTION 53 — CALIBRATION CONSTANTS PENDING TC-02

Constant	Current Est.	Source
DESCENT_CTRL_HOVER_FREQ_HZ	2.50 Hz	TC-02-03: load cell reads zero
DESCENT_CTRL_K_FREQ_PER_FPM	0.0009	TC-02-07: from hover freq and slope
ACTIVE_DAMP_K_AMP	0.40	TC-04-05: free flight roll response
H-26 behaviour	Unknown	TC-02-06: ROLL or LATERAL_FORCE
_H35_V_CRUISE_ENTRY	5.0 m/s	TC-03: first lift-wing transition speed
_H35_V_CRUISE	8.0 m/s	TC-03: 2-pair lift configuration speed

SECTION 54 — PRE-FIRST-FLIGHT CHECKLIST SUMMARY

Done	Item	Notes
✗	mco_loadcell.py --calibrate	1 kg mass. Required before TC-01.
✗	mco_build_measurements.py	15 physical measurements. Required before TC-01.
✗	mco_odrive_setup.py	Configure all 8 S1 nodes.
✗	mco_odrive_setup.py --check-magnet	All 8 magnets status 1.
✗	mco_odrive_setup.py --calibrate	Motor calibration (no encoder cal needed).
✗	mco_rc_verify.py	CH1=Roll, CH2=Pitch, CH3=Lever, CH4=Yaw.
✗	mco_preflight_checks.py --bench	All 19 checks pass (inc. 5 S1-specific).
✗	mco_vibration_check.py --quick	Baseline before first motor run.
✗	TC-01: tc01_start	16 steps bench test.
✗	mco_vibration_check.py (full)	After TC-01-06, before TC-01-07.
✗	mco_log_analysis.py --tc01	Review thrust curve.
✗	TC-02: tc02_start	8 steps captive hover.

✗	mco_log_analysis.py --tc02	Confirm hover frequency extracted.
✗	TC-03: tc03_start	6 steps truck forward flight.
✗	CT-J curve confirmed	Required before H-33 State C / H-35 2-pair / H-36 stow.
✗	Pitot reading verified	V_ms non-zero when truck moving. Correct src: msg.airspeed.
✗	TC-04: tc04_start	7 steps — first free flight.
✗	Post TC-04 motor health	python3 mco_log_analysis.py and last_flight_health.json.

SUPPLEMENT TO PART I — PILOT INTENT CONTROL SYSTEM (Revision 17)

This section describes the changes to how the pilot controls the aircraft. These changes are in effect from TC-04 first free flight onwards.

SECTION S1 — CONTROL CONVENTION

■ WARNING: The control convention has changed from the previous design. Read this section carefully before TC-04.

Input	Axis	Effect
JOYSTICK left/right	Roll	Banks the aircraft left or right. Identical to a conventional fixed-wing aircraft.
JOYSTICK fore/aft	Pitch	Forward stick = nose down / accelerate. Aft stick = nose up / climb. Identical to fixed-wing.
PEDALS left/right	Yaw	Left pedal = nose left. Right pedal = nose right. Identical to fixed-wing rudder pedals in every respect.
LEVER forward (100%)	Full thrust	Maximum flapping frequency. Maximum climb rate. Identical to pushing the throttle forward on a fixed-wing aircraft.
LEVER mid (50%)	Trim / hold	Aircraft holds current altitude. Equivalent to cruise power in a fixed-wing aircraft.
LEVER aft (0%)	Airbrake / glide	Zero thrust. Maximum drag. Wings enter glide configuration. Equivalent to idle + airbrake on a fixed-wing aircraft.

✓ The lever convention is the most important change. In the previous design, lever aft was maximum thrust. It is now lever forward = maximum thrust. This matches every fixed-wing aircraft. Verify with mco_rc_verify.py before TC-04 — it now confirms the forward=thrust direction explicitly.

● CAUTION: The RC transmitter lever channel (CH3) must be configured so that pushing the lever forward produces a higher value (forward = more power). mco_rc_verify.py checks this and will flag a REVERSED warning if wrong.

SECTION S2 — ALL MECHANISMS AVAILABLE TO ALL AXES

The pilot communicates only intent — roll left, pitch up, yaw right. The software decides which combination of wing mechanisms executes that intent most effectively. The optimizer learns which mechanisms work best on this specific aircraft through flight experience.

Five mechanisms are available for each axis:

Axis	Mechanism	Description
ROLL	roll_amp	Amplitude differential — port wings flap higher than stbd or vice versa. Available only when wings are flapping. Fastest response.

ROLL	roll_sweep	Sweep asymmetry — inner-turn wings swept aft, outer-turn wings swept forward. Available in both powered and glide flight.
ROLL	roll_aoa	Per-wing AoA differential (H-25) — port wings at different pitch angle from stbd wings. Four times faster than amplitude. Available always.
ROLL	roll_phase	Cascade phase asymmetry — port side phase advanced relative to stbd. Only in powered flight. Creates timing-based lift differential.
ROLL	roll_dih	Dihedral asymmetry — port wings tilted up, stbd down (or reverse). Available always. Redirects lift vector.
PITCH	pitch_freq	Frequency change — all wings flap faster or slower collectively. Primary pitch mechanism in powered flight. Controls climb/descent rate.
PITCH	pitch_aoa	Collective AoA — all eight wings increase or decrease pitch angle together. Primary pitch mechanism in glide.
PITCH	pitch_sweep	Collective sweep — all wings sweep forward (accelerate) or aft (decelerate). Changes stroke plane tilt.
PITCH	pitch_phase	Cascade phase advance — all pairs phase-shifted together. Biases the thrust vector forward or rearward.
PITCH	pitch_dih	Collective dihedral — all wings tilted up together. Analogous to flare. Redirects lift forward.
YAW	yaw_phase	Cascade phase asymmetry — port side advanced relative to stbd. Primary yaw mechanism in powered flight. Differential thrust moment.
YAW	yaw_dih	Dihedral asymmetry — port up, stbd down or vice versa. Redirects thrust vector sideways. Always available.
YAW	yaw_amp	Amplitude asymmetry with opposing AoA — torque-based yaw. Creates a yawing moment through differential flapping torque.
YAW	yaw_sweep	Differential sweep — one side more aft sweep = more drag = yaws toward that side. Available in both powered and glide.
YAW	yaw_h26	Lateral force mode — only available if TC-02 classifies the aircraft as H-26 LATERAL_FORCE. Generates sideways force.

SECTION S3 — HOW THE OPTIMIZER LEARNS WHICH MECHANISMS WORK

All 15 mechanisms start with equal weight (0.200 each within each axis). After every tick where the pilot made a meaningful input, the optimizer compares the achieved rotation rate (from IMU) against the expected rate for that input. Mechanisms that were active when the response was good have their weight increased. Mechanisms active when response was poor have their weight decreased. The sum of weights per axis is always 1.0.

Over many flights, the weight vector converges to favour whichever combination of mechanisms is most effective on the actual physical aircraft — not on the theoretical model. This matters because the effectiveness of each mechanism depends on the actual servo bandwidth, rope drive stiffness, wing mass distribution, and aerodynamic coupling effects that can only be measured by flying.

Ground Station Command	Purpose
mechanism_weight_report	Shows current weight ranking for all 15 mechanisms across three axes. The highest-weighted mechanism in each axis is the dominant one.

save_mechanism_weights	Saves current weights to /home/pi/knowledge/mechanism_weights.json. Run after every productive flight session. Weights are also saved automatically at shutdown.
load_mechanism_weights	Loads previously saved weights. Run automatically at startup. Resets learning to the previously converged state.
set_h26_classification ROLL	Set after TC-02-06 result. Activates the appropriate yaw_h26 behaviour. Use: set_h26_classification LATERAL_FORCE if TC-02 classifies the aircraft as having sidestep capability.
mechanism_weight_reset	Resets all weights to equal (0.200). Use if converged to a bad local minimum or after major hardware changes.

SECTION S4 — LEVER ZONE DEFINITIONS

Lever %	Zone	What happens
100%	Full thrust / Sprint	Maximum flapping frequency. H-33/35 thrust wings at full freq. Use for: maximum climb rate, acceleration, emergency go-around.
75–99%	Climb power	High frequency. Aircraft climbs at commanded pitch attitude. Equivalent to climb power on a fixed-wing aircraft.
55–75%	Cruise power	Balanced frequency. Aircraft maintains altitude at cruise speed. Optimizer explores this zone during Discovery Mode.
45–55%	Trim / altitude hold	Auto-hold altitude. 50% is the exact trim detent. ±5% hysteresis band. Equivalent to cruise trim on a fixed-wing.
25–45%	Approach / descent	Reduced frequency. Aircraft descends at commanded pitch. Equivalent to reduced power on approach.
10–25%	Glide / soar	Wings transition to fixed-wing glide configuration. Best glide at ~20%. Equivalent to idle + flaps up.
0–10%	Maximum airbrake	Wings at maximum drag configuration. Maximum sink rate. Use for: steep approach, energy dumping, emergency descent. Equivalent to idle + full airbrake on a fixed-wing aircraft.

SECTION S5 — GLIDE AND UNPOWERED FLIGHT

When the lever is reduced below approximately 25%, the wings progressively transition from flapping to a fixed-wing glide. In the pure glide (lever below 10%):

Element	Glide behaviour
All eight wings	Held stationary at best-lift-to-drag AoA. Zero motor power. Fixed-wing aerofoils in the airstream.
Roll control	Sweep asymmetry (roll_sweep) and AoA differential (roll_aoa) are available. Amplitude differential is unavailable (no flapping). The optimizer will have learned which mechanisms remain effective by the time glide is used extensively.
Pitch control	Collective AoA (pitch_aoa) is primary. More AoA = slower / shallower glide. Less AoA = faster / steeper glide. Frequency mechanism unavailable (wings stationary).

Yaw control	Differential sweep (yaw_sweep) and dihedral asymmetry (yaw_dih) are available. Phase mechanism unavailable (no flapping). Coordinated turns via banked glide — identical to a sailplane.
H-33/34/35/36	Wing morphing hypotheses remain active in glide. Stopped pairs (State B) are essentially in glide already. The H-35 lift wings are fixed-wing surfaces. In a full glide H-35 and H-36 configurations merge with the glide configuration.

SECTION S6 — TC-04 SPECIFIC NOTES

TC-04 step 5 (roll response) and step 6 (forward flight) will provide the first real data for the mechanism weight optimizer. Mechanism weights are saved automatically at shutdown after TC-04.

Step	Action	Expected / Notes
TC-04-05a	Verify roll convention before manoeuvre	Apply small right roll stick input. Aircraft should bank RIGHT. If it banks left: transmitter roll channel is reversed. Stop immediately and reverse CH1 in transmitter.
TC-04-05b	Verify pitch convention	Apply small forward pitch stick input. Aircraft should pitch nose DOWN / accelerate slightly. If it pitches up: reverse CH2 in transmitter.
TC-04-05c	Verify yaw convention	Apply small right pedal (right rudder). Aircraft nose should move RIGHT. If it moves left: reverse CH4 in transmitter.
TC-04-05d	Verify lever convention	Advance lever forward from 50% to 70%. Aircraft should climb or increase frequency. If frequency decreases: lever is reversed. Reverse CH3 in transmitter immediately.
TC-04-07	Check mechanism weights after discovery	After TC-04-07 (5-minute free flight discovery), type: mechanism_weight_report on ground station. This shows which mechanisms the optimizer is starting to favour based on the first flight responses. Save with: save_mechanism_weights

✓ Technical: lever_internal() in mco_wave_library.py converts the fixed-wing lever value (0=idle, 100=thrust) to the legacy internal format (0=sprint, 50=trim, 100=airbrake) used by the pre-existing power management layer. This conversion is transparent — pilots only see the fixed-wing convention.